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Fourier Transformation of a function in Continuous domain

↳ Impulse shifting.

$$\mathcal{F}\{f(t)\} = -\int_{-\infty}^{\infty} f(t)\, e^{-j2\pi ut}\, dt$$

$F(u)$ ⇓ ↓ u

$$\int_{-\infty}^{\infty} f(t)\left[\cos(2\pi ut - ut) - j\sin(2\pi ut)\, dt\right)$$

f.T

To Calculate ~~unit impulse~~ of ~~fourier transform~~ unit impulse located at 0

$f(t) =$ ∮ unit impulse located at 0

$f(t) = 1$ at $t = 0$

fourier transformation would be invertible.

↳ of unit impulse = 1.

Apply Shifting property → located at $t = t_0$

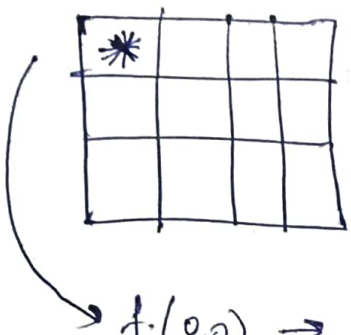
$$\int_{-\infty}^{\infty} f(t - t_0)\, e^{-j2\pi ut}\, dt$$

$$\int_{-\infty}^{\infty} f(t+t_0)\, e^{-j2\pi ut}\, dt$$

↳ f.T is location variant.

DC Components

Pixel at 0,0 position of an image is the DC component.





→ Mean

$$\text{Mean} = \frac{9}{3} = 1$$

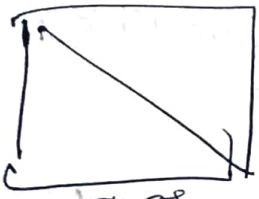
$f(0,0) \rightarrow f(0,0)$

1 = DC component

DCT, DST
↓ ↓ → transform
Cosine Sine

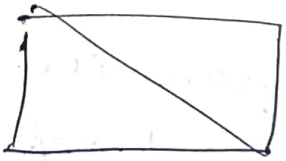
Discrete.

DC component in image. = fourier transform of $f(0,0)$
lowest.



DC

highest to lowest



DST

highest

Convolution in frequency domain:

↳ Gonzales

↳ fourier tr. of two functions in Spatial domain = fourier transform in individual functions.

$$f(t) * h(t) = \int_{-\infty}^{\infty} f(z)\, h(t-z)\, dz$$

F.T

$$\mathcal{F}[f(t) * h(t)] = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(z) h(t-z) dz \right] e^{-i2\pi st}\, dt$$

#

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$x = \{1, 2, 8, 7\}$ → 1d vector

$y = (\text{Kernel})x$

↓

w

$w_M = \exp\left(\frac{-j2\pi}{M}\right)$

↳ nth root of unity

$M = 5$

$$W_5 = \begin{pmatrix} W_M^0 & W_M^0 & W_M^0 & W_M^0 & W_M^0 \\ W_M^0 & W_M^1 & W_M^2 & W_M^3 & W_M^4 \\ W_M^0 & W_M^2 & W_M^4 & W_M^6 & W_M^8 \\ W_M^0 & W_M^3 & W_M^6 & W_M^9 & W_M^{12} \\ W_M^0 & W_M^4 & W_M^8 & W_M^{12} & W_M^{16} \end{pmatrix}$$

$x = \{1, 2, 8, 9\}$

$M = 4$

$$W = \exp\left(\frac{-2\pi j}{4}\right) = \exp\left[\frac{-\pi j}{2}\right]$$

$$= \cos\left(\frac{-\pi}{2}\right) + j \sin\left(\frac{-\pi}{2}\right)$$

$$= -j \quad (\sin(-\theta) = -\sin\theta)$$

$$W_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & (-j)^2 & (-j)^3 \\ 1 & (-j)^2 & (-j)^4 & (-j)^6 \\ 1 & (-j)^3 & (-j)^6 & (-j)^9 \end{bmatrix} \times \begin{bmatrix} 1 \\ 2 \\ 8 \\ 9 \end{bmatrix}$$

$$W_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & +j \\ 1 & -1 & 1 & -1 \\ 1 & +j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 8 \\ 9 \end{bmatrix}$$

$$W_4 = \begin{bmatrix} 1+2+8+9 \\ 1-2j-8+9j \\ 1-2+8-9 \\ 1+2j-8-9j \end{bmatrix} = \begin{bmatrix} 20 \\ -7+7j \\ -2 \\ -7-7j \end{bmatrix}$$

Ques $f(y) = \frac{1}{N}[W^{-1}] \times f(x)$

W^{-1} = Hermitian matrix where sign of exponential function is +ve.

$$\therefore W^{-1} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & +j \end{bmatrix}$$

→ just conjugate of the matrix.

$$W^{-1}\begin{pmatrix}20\\-7+7j\\-2\\-7-7j\end{pmatrix} = \frac{1}{4}\begin{bmatrix}4\\8\\32\\36\end{bmatrix} = \begin{bmatrix}1\\2\\8\\9\end{bmatrix}$$

$$\frac{1}{4}\begin{bmatrix}1 & 1 & 1 & 1\\ 1 & j & -1 & -j\\ 1 & -1 & 1 & -1\\ 1 & -j & -1 & j\end{bmatrix}\begin{bmatrix}20\\-7+7j\\-2\\-7-7j\end{bmatrix} = \begin{bmatrix}20-7+7j-2-7-7j\\20-7j-7+2-7j\\20+7-7j-2+7+7j\\20-7j+7+2-7j\end{bmatrix}\begin{matrix}4\\8\\32\\36\end{matrix}$$

$$= \begin{bmatrix}4\end{bmatrix}$$

Ques: Proof that DfT works on given sample if it works to inverse nikaalne ka same aayega, otherwise same nhi aayega

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Properties of DfT

↳ gonzales

and their Proof

for both 1D and 2D

translation, rotation, commutative, associative.

Table 4.1 → Proof questions

DCT, DST; Compression → Arithmetic
RLE
Lempel Ziv
Huffman

↳ Ratio of Compression

DCT :-

$$DCT \rightarrow f(u) = \sqrt{2}\,\frac{w(x)}{\sqrt{N}} \sum_{x=0}^{n-1} f(x) \cos\left(\frac{(2x+1)u\pi}{2N}\right)$$

where $u = 0, 1, 2, \ldots N-1$

Here $w(u) = \begin{cases} \frac{1}{\sqrt{2}} & \text{if } u=0 \\ 0 & \text{if } u \neq 0 \end{cases}$

Inverse :-

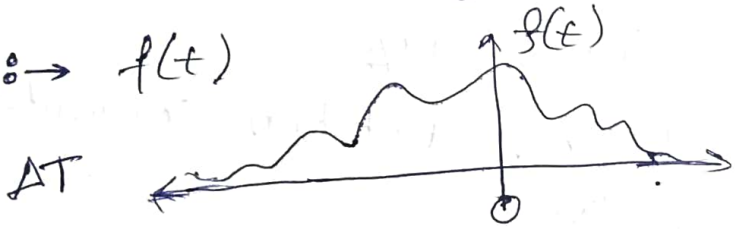
$$f(u) = \sqrt{\frac{2}{N}} \sum_{u=0}^{N-1} w(u) \times f(u) \times \cos\left(\frac{2uf1\ u\pi}{2N}\right)$$

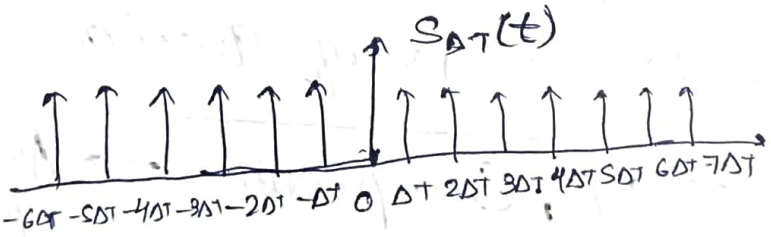
when $w(u) = \begin{cases} \frac{1}{\sqrt{2}} & \text{if } u=0 \\ 1 & \text{if } u \neq 0 \text{ for all } x = 0, \ldots N-1. \end{cases}$

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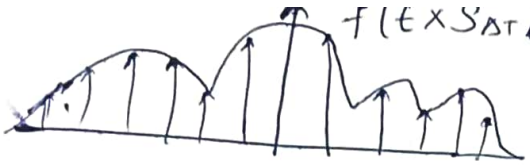
Fourier transform of Sampled function

Sampling :→ $f(t)$





→ Multiply $f(t)$ by Sampling
↳ function equal to a train of impulses Δ T units apart



output of multiplying both

$$\hat{f}(t) = f(t) \cdot S_{\Delta T}(t)$$

→ on sampling exact replication doesn't happen.

$$= \sum_{-\infty}^{\infty} f(t) \cdot f(t - n\Delta T)$$

$$f_k = \int_{-\infty}^{\infty} f(t)\, \delta(t - k\Delta T)\, dt \quad -(1)$$

$$= f(k\Delta T)$$

$$\hat{F}(u) = \mathcal{F}\{\hat{f}(t)\} = \mathcal{F}\{f(t) \times S_{\Delta T}(t)\}$$

$$\boxed{\hat{F}(u) = F(u) * S(u)}$$

↓ Original function Fourier

↓ Sampled function Fourier

Fourier transform of 2D functions

↳ C language → DIP + C → 10 marks
↳ time space complexity.

$$f(u,v) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u,v)\, e^{j2\pi \left(\frac{uv}{M} + \frac{vu}{N}\right)}$$

Proof of Convolution →

Fourier transform of convolution of two functions in spatial domain is equal to product of the frequency domain of the F.T of two functions

Homo-morphic filters

- # High Pass / Low Pass → Ideal High Pass
- # Band Pass
- # Gaussian High Pass
- # Butterworth high Pass
- #

} frequency domain

$$H_H(P,Q) = 1 - H_L(P,Q)$$

Spatial domain → high pass and low pass are mirror images of each other.

Images → to be identified by seeing filter → which filter is applied

Homomorphic filters

↳

Image = $i * R$

$f(x,y) = i(x,y) \cdot r(x,y)$

fourier of $f \neq$ fourier of $i(x,y)$ · fourier of $r(x,y)$

$$F.T\ \mathcal{F}\{f(x,y)\} \neq \mathcal{F}[i(x,y)]\,\mathcal{F}[r(x,y)] \quad -(1)$$

Suppose we define - $Z(u,v) = \ln f(x,y)$

$$= \ln [i(u,v) \cdot r(x,y)]$$

$$= \ln (i(x,y) + \ln r(x,y))$$

$-(2)$

Then,

$$\mathcal{F}[z(x,y)] = \mathcal{F}(\ln f(x,y)]$$

$$= \mathcal{F}\{\ln i(x,y)\} + \mathcal{F}\{\ln r(x,y)\} \quad (3)$$

where $= \boxed{f_i(x,y) + f_r(x,y)}$

$F_i(u,v)$ & $f_r(u,v)$ are

F.T of I and r

$z(u,v)$

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Image compression

↳ method of compression is same for all.

↳ Coding Redundancy

⇒ Numerical

Ques:- An image Scan centre needs to store medical images whose resolution is $1024 \times 1024 \times 24$ bits. A total of 10,000 images are present how much storage & transmission time will they require at 64 KBPS

Source (Type of data)
- ↳ Image
- ↳ Video
- ↳ Text
- ↳ Binary
- ↳ Sound
- ↳ Graphics

$C = \frac{b'}{b}$ → compressed / → original

Sir's → $\frac{\text{original}}{\text{compressed}}$

$$\boxed{R = 1 - \frac{1}{C}}$$

↳ Relative data Redundancy

Saving percentage :- 1 - Message Size after comp / Message (code) Size before Compression.

Bitrate = Size of Compressed file / Notal no of pixels in image.

Types of Compression

↳ ① Lossy compression
② Lossless Compression.



$-\log_2 P_i$
interpixel
irrelevant data → four types of redundancy.

lossy
↳ removal of redundant data in a controlled way.

↳ lossless
↳ only we change the way of representation of coding symbols
↳ irreversible.

$\sum L \times P(x_i)$

Arithmetic Coding
Average Codeword length.
Huffman Coding → unique.

Run length Coding

Coding Redundancy = Avg length - Entropy

$$\text{Entropy} = -P_i \log_2 P_i$$

Huffman Coding → lossless

$$\eta = \frac{\text{Entropy}}{\bar{R}}$$

$\bar{R}$ → avg codelength

↳ level of image.

↳ follow a single Pattern of Coding

↳ [redacted]

↳ drawbacks

$$\text{efficiency} = \frac{\text{Entropy}}{\bar{R}}$$

$P(x_1) = 0.37$
$P(x_2) = 0.33$
$P(x_3) = 0.16$
$P(x_4) = 0.07$
$P(x_5) = 0.04$
$P(x_6) = 0.02$
$P(x_7) = 0.01$

0.03 → 0.07 → 0.14 → 0.30 → 0.63 → 1

↳ Gonzales Book convention.

$x_1 = 0$
$x_2 = 10$
$x_3 = 110$
$x_4 = 1110$
$x_5 = 11110$
$x_6 = 111110$
$x_7 = 111111$

$$\bar{R} = 0.37 \times 1 + 2 \times 0.33 + 3 \times 0.16 + 4 \times 0.07 + 5 \times 0.04 + 6 \times 0.02 + 6 \times 0.01$$

$$\bar{R} = 0.37 + 0.66 + 0.48 + 0.28 + 0.20 + 0.12 + 0.06$$

$$= 2.17$$

$$\text{Entropy} = -P \log P$$

$$= -(0.37 \log(0.37 + 0.33 \log_2(0.33) + 0.16 \log_2(0.16) + 0.07 \log_2(0.07) + 0.04 \log_2(0.04) + 0.02 \log_2(0.02) + 0.01 \log_2(0.01))$$

$$= \frac{2.115}{2.17} = 0.9746$$

$$= 97.46\ \%$$

Drawbacks

↳ not unique code → for same efficiency
↳ not unique
↳ no one to one mapping → as we can decide which side is 0, which side is 1

Advantages

↳ optimal

Arithmetic Coding

0.05 0.3

↳ Pattern ki coding

Drawback

↳ Pattern coding

↳ we always get a range → which can represent that pattern.

↳ no uniqueness

↳

RLE

7 → 12 bits

101100 → (01,1)(01,0)(10,0)(10,0)

↓

2 bits for count

Bit Plane Coding (Bit Plane Slicing)

Lempel-ziv Coding.